

PhD Math Camp

Limits and Continuity

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Sequences and Series

Sequences

- A sequence is an ordered list of numbers.
- Example sequences:
 - $\{3, 7, 5, 6, 8, \dots\}$
 - $\{x_n\}_{n=1}^N = \{n + 1\}$
- All sequences are countable, meaning that one can assign a natural number to each element in the sequence.
- Notation:
$$\{x_n\}_{n=1}^N = \{x_1, x_2, x_3, \dots, x_N\}$$
- A sequence can be finite or infinite.

Series

- A series is the sum of a sequence.
- **Example.** Consider the series $A_n = \{n\}$, then the following is the series of the first 10 terms of the sequence:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

- Series notation:

$$\sum_{i=1}^N x_i$$

- In this case, we used $\sum_{n=1}^{10} n$ to compute the sum of the first 10 natural numbers, which equals 55. This can be calculated using the formula for the sum of the first N natural numbers:
$$\sum_{k=1}^N k = \frac{N(N+1)}{2}.$$

Geometric Series

- Geometric series: The sum of a sequence that has a constant ratio between successive terms.
 - Example: $1 + 2 + 4 + 8 + \dots$
 - General form:

$$a + ar + ar^2 + \dots = \sum_{k=0}^{\infty} ar^k$$

- Convergence of geometric series:
 - If $|r| < 1$ and there are infinite terms, the series converges to the sum:

$$S_{\infty} = \frac{a}{1 - r}$$

- For any $r \neq 1$ with finitely many terms n , the sum is:

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

Limits of Sequences

Convergence and Divergence

- A limit describes the behavior of a function, sequence, or series of numbers as it approaches a given value.
- A sequence converges if it has a finite limit.
- A sequence diverges if it either:
 - Has no limit, or
 - Has a limit of $\pm\infty$.

Convergence

Example sequence: $\{A_n\} = 2 - \frac{1}{n^2}$

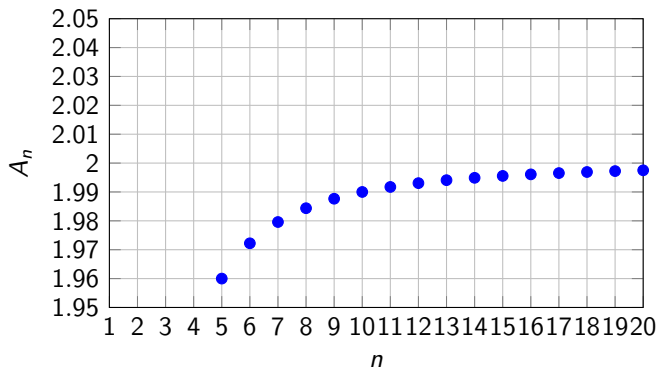


Figure 1: $A_n = 2 - \frac{1}{n^2}$

Divergence

Example sequence: $\{B_n\} = n + 1$

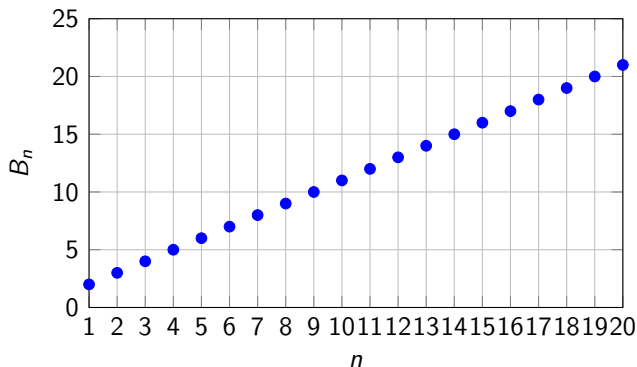


Figure 2: $B_n = n + 1$

Limits of a Sequence: Formal Definition

The ϵ - N definition

The sequence $\{x_n\}$ has the limit L , which we write as:

$$\lim_{n \rightarrow \infty} x_n = L$$

if for any $\epsilon > 0$ there is an integer N with the property that $|x_n - L| < \epsilon$ for each $n > N$.

Limits of a Sequence: A Worked Proof

Consider the sequence $\{A_n\} = 2 - \frac{1}{n^2}$. We claim that:

$$\lim_{n \rightarrow \infty} A_n = 2$$

Proof. Let $\epsilon > 0$ be given. We need to find N such that if $n > N$, then $|A_n - 2| < \epsilon$.

$$|A_n - 2| = \left| 2 - \frac{1}{n^2} - 2 \right| = \frac{1}{n^2}$$

We want $\frac{1}{n^2} < \epsilon$. Solving for n , we get:

$$n > \frac{1}{\sqrt{\epsilon}}$$

Therefore, we can choose N to be any integer greater than $\frac{1}{\sqrt{\epsilon}}$. Thus, for all $n > N$,

$$|A_n - 2| < \epsilon$$

which confirms our claim that $\lim_{n \rightarrow \infty} A_n = 2$. ■

Properties of Limits of Sequences

Let $\lim_{n \rightarrow \infty} y_n = y$ and $\lim_{n \rightarrow \infty} z_n = z$. Then:

1. $\lim_{n \rightarrow \infty} [ky_n + \ell z_n] = ky + \ell z$
2. $\lim_{n \rightarrow \infty} y_n z_n = yz$
3. $\lim_{n \rightarrow \infty} \frac{y_n}{z_n} = \frac{y}{z}$, if $z \neq 0$

Example.

Find the limit of $\lim_{n \rightarrow \infty} \frac{n+3}{n}$

Limits of Functions

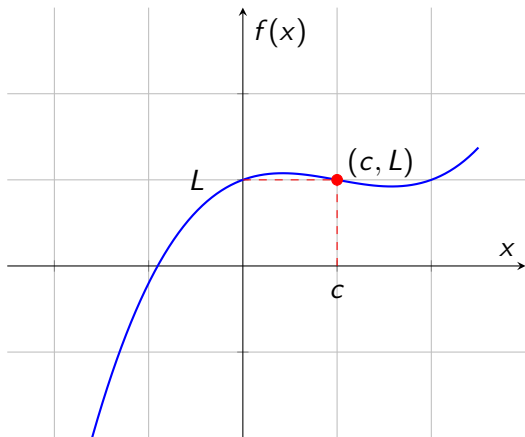
Limits and Functions

- For the function $f(x)$, a limit is the value that $f(x)$ tends toward given arbitrarily small movements toward a specific value of x , say $x = c$.
- Denoted as:

$$\lim_{x \rightarrow c} f(x) = L$$

- The limit of $f(x)$ as x approaches c is L .

Visualizing the Limit of a Function



$$\lim_{x \rightarrow c} f(x) = L$$

One-Sided Limits

- It's possible to approach a point from two different directions:
 - From the right (approaching from values greater than c):

$$\lim_{x \rightarrow c^+} f(x) = L^+$$

- From the left (approaching from values less than c):

$$\lim_{x \rightarrow c^-} f(x) = L^-$$

- If the limits from above and below are equal, then the function has a unique limit at c .
- The existence of a limit depends on the behavior of the function near a specific value of the variable(s), and this is not necessarily the same as the value of the function at that specific value.

Limit of a Function: Formal Definition

The ϵ - δ definition

Let $f(x)$ be defined at each point in some open interval containing the point c , except possibly at c itself. We say:

$$\lim_{x \rightarrow c} f(x) = L$$

if for any $\epsilon > 0$, there exists a real $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

Delta-Epsilon: A Worked Proof

Problem. Prove that $\lim_{x \rightarrow 2}(3x - 5) = 1$ using the delta-epsilon definition.

Prop. For every $\epsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - 2| < \delta$, then $|(3x - 5) - 1| < \epsilon$.

Direct proof.

- Start with the inequality: $|(3x - 5) - 1| = |3x - 6| = 3|x - 2|$.
- We want $3|x - 2| < \epsilon$, so divide by 3: $|x - 2| < \frac{\epsilon}{3}$.
- Therefore, choose $\delta = \frac{\epsilon}{3}$.
- For every $\epsilon > 0$, if $0 < |x - 2| < \delta = \frac{\epsilon}{3}$, then $|(3x - 5) - 1| < \epsilon$. This satisfies the delta-epsilon definition, proving that $\lim_{x \rightarrow 2}(3x - 5) = 1$.



Properties of Limits

Let $f(x)$ and $g(x)$ be functions such that the following limits exist:

$$\lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = M$$

Then, the following properties hold:

- **Limit of a Sum:**

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + M$$

- **Limit of a Difference:**

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L - M$$

- **Limit of a Product:**

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M$$

- **Limit of a Quotient:**

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}, \quad \text{provided } M \neq 0$$

- **Limit of a Composition:** $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(M)$, provided f is continuous at M

Squeeze Theorem

Theorem

If $f(x) \leq g(x) \leq h(x)$ for every x not equal to c and

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = L$$

then

$$\lim_{x \rightarrow c} g(x) = L$$

Examples

Here are some examples of computing limits:

1. $\lim_{h \rightarrow 0} \left[\frac{2ah + h^2}{h} \right]$

2. $\lim_{x \rightarrow 3} \left[\frac{x^2 + 2x - 15}{x - 3} \right]$

3. $\lim_{x \rightarrow \infty} \left[\frac{5x^3 - 2x^2 + x + 7}{3x^3 + x^2 - 4} \right]$

4. $\lim_{x \rightarrow 1} \left[\frac{3}{x - 1} \right]$

Solutions to the Limit Examples

$$2. \lim_{x \rightarrow 3} \left[\frac{x^2 + 2x - 15}{x - 3} \right]$$

Solution.

Factor the numerator:

$$x^2 + 2x - 15 = (x - 3)(x + 5).$$

Simplify the expression:

$$\frac{(x-3)(x+5)}{x-3} = x + 5, \text{ for } x \neq 3.$$

Apply the limit:

$$\lim_{x \rightarrow 3} (x + 5) = 3 + 5 = 8.$$

$$3. \lim_{x \rightarrow \infty} \left[\frac{5x^3 - 2x^2 + x + 7}{3x^3 + x^2 - 4} \right]$$

Solution.

Divide numerator and

denominator by x^3 : $\frac{5x^3 - 2x^2 + x + 7}{x^3}$

and $\frac{3x^3 + x^2 - 4}{x^3}$.

This simplifies to $\frac{5 - \frac{2}{x} + \frac{1}{x^2} + \frac{7}{x^3}}{3 + \frac{1}{x} - \frac{4}{x^3}}$.

As $x \rightarrow \infty$, the terms with $\frac{1}{x}$, $\frac{1}{x^2}$, and $\frac{1}{x^3}$ approach 0, so the limit is

$$\frac{5-0+0+0}{3+0-0} = \frac{5}{3}.$$

Plot of $f(x) = \frac{3}{x-1}$ with Vertical Asymptote at $x = 1$

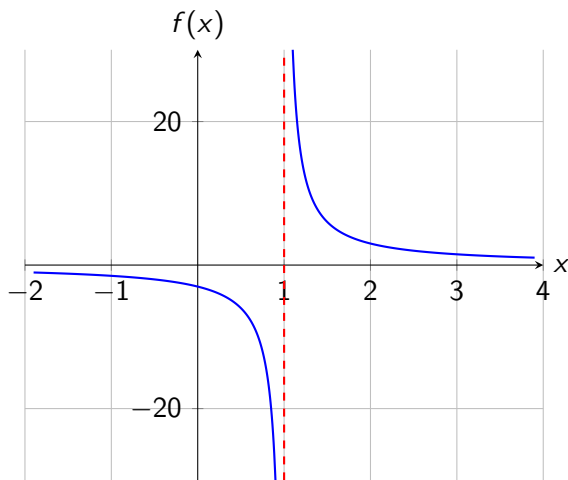


Figure: $f(x) = \frac{3}{x-1}$, with the vertical asymptote at $x = 1$ dashed.
The two branches are drawn separately because the function is undefined there.

Applications

Law of Large Numbers: An Application of Limits

The Law of Large Numbers (LLN) is a fundamental theorem in probability theory that describes the result of performing the same experiment a large number of times. It shows how the sample mean converges to the expected value as the sample size increases.

Thm. (Weak Law of Large Numbers) Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with expected value $\mathbb{E}[X_i] = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2$. Then, for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| < \epsilon \right) = 1$$

Interpretation. As the sample size n increases, the sample mean $\frac{1}{n} \sum_{i=1}^n X_i$ converges in probability to the true mean μ .

Condorcet Jury Theorem: An Application of Limits

Consider a group of N voters (N odd) deciding between two alternatives, A and B. Each voter independently chooses the correct alternative with probability $p > 0.5$.

- **Individual Probability of Correctness:** Each voter has a probability p of choosing the correct alternative.
- **Group Decision:** The group uses majority rule to decide between the two alternatives.

The probability P_N that the majority vote is correct is given by:

$$P_N = \sum_{k=\lceil N/2 \rceil}^N \binom{N}{k} p^k (1-p)^{N-k}$$

Limit Result:

$$\lim_{N \rightarrow \infty} P_N = 1 \quad \text{if} \quad p > 0.5$$

Continuity

Continuity

Intuitively, a continuous function is a function without sudden breaks in it. (Moore and Segal)

Def. Suppose that the domain of the function f includes an open interval containing the point c . f is continuous at c if:

1. $\lim_{x \rightarrow c} f(x)$ exists, and
2. $\lim_{x \rightarrow c} f(x) = f(c)$.

Examples of Continuous and Discontinuous Functions

Continuous Function: $f(x) = x^2$ at $c = 2$

- $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} x^2 = 4$
- $f(2) = 4$
- Both conditions are satisfied, so $f(x)$ is continuous at $x = 2$.

Discontinuous Function: $g(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$ at $c = 1$

- $\lim_{x \rightarrow 1} g(x) = 2$ (as $g(x) = x + 1$ when $x \neq 1$)
- $g(1) = 0$
- $\lim_{x \rightarrow 1} g(x) \neq g(1)$, so $g(x)$ is discontinuous at $x = 1$.

Discontinuous Function Example

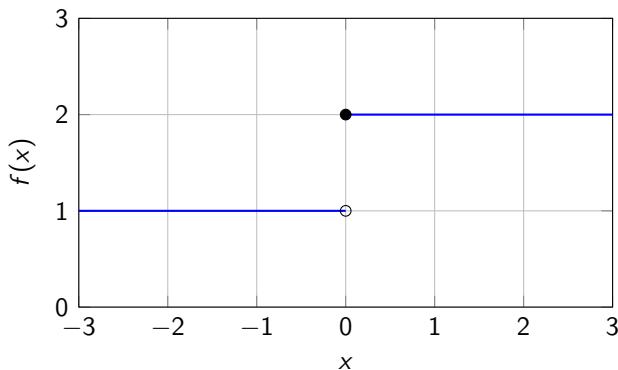


Figure: A discontinuous function defined as $f(x) = 1$ for $x < 0$ and $f(x) = 2$ for $x \geq 0$.

Continuity of the Absolute Value Function

Absolute Value Function:

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Continuity Check:

- For $x > 0$, $f(x) = x$, which is continuous.
- For $x < 0$, $f(x) = -x$, which is continuous.
- At $x = 0$:

$$f(0) = |0| = 0,$$

$$\lim_{x \rightarrow 0^-} f(x) = 0,$$

$$\lim_{x \rightarrow 0^+} f(x) = 0,$$

$$\text{Thus, } \lim_{x \rightarrow 0} f(x) = f(0) = 0.$$

Conclusion: The absolute value function $f(x) = |x|$ is continuous for all $x \in \mathbb{R}$.

Intermediate Value Theorem

Thm. If a function $f(x)$ is continuous on the closed interval $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there exists at least one c in the interval (a, b) such that $f(c) = 0$.

Example.

Consider $f(x) = x^3 - 6x + 2$ on the interval $[0, 1]$.

- $f(0) = 2$, $f(1) = -3$
- Since $f(x)$ is continuous and $f(0)$ and $f(1)$ have opposite signs, by the IVT, there exists some $c \in (0, 1)$ such that $f(c) = 0$.

Extreme Value Theorem

Thm. If a function $f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ attains both a maximum and a minimum value on that interval. In other words, there exist points $c, d \in [a, b]$ such that:

$$f(c) \geq f(x) \geq f(d) \quad \text{for all } x \in [a, b].$$

Example.

Consider $f(x) = -x^2 + 4x + 1$ on the interval $[0, 3]$.

- $f(x)$ is a continuous function.
- Hence, there exist points $c, d \in [0, 3]$ such that:

$$f(c) \geq -x^2 + 4x + 1 \geq f(d) \quad \text{for all } x \in [0, 3].$$

Importance of IVT and EVT in Statistics

Intermediate Value Theorem (IVT):

- Ensures that if a continuous function changes sign over an interval, there exists a root within that interval.
- Used in numerical methods for finding roots, essential in optimization and solving statistical equations.

Extreme Value Theorem (EVT):

- Guarantees that a continuous function on a closed interval has both a maximum and minimum value.
- Crucial for optimization problems, where finding the maximum likelihood or minimum variance involves identifying global optima.